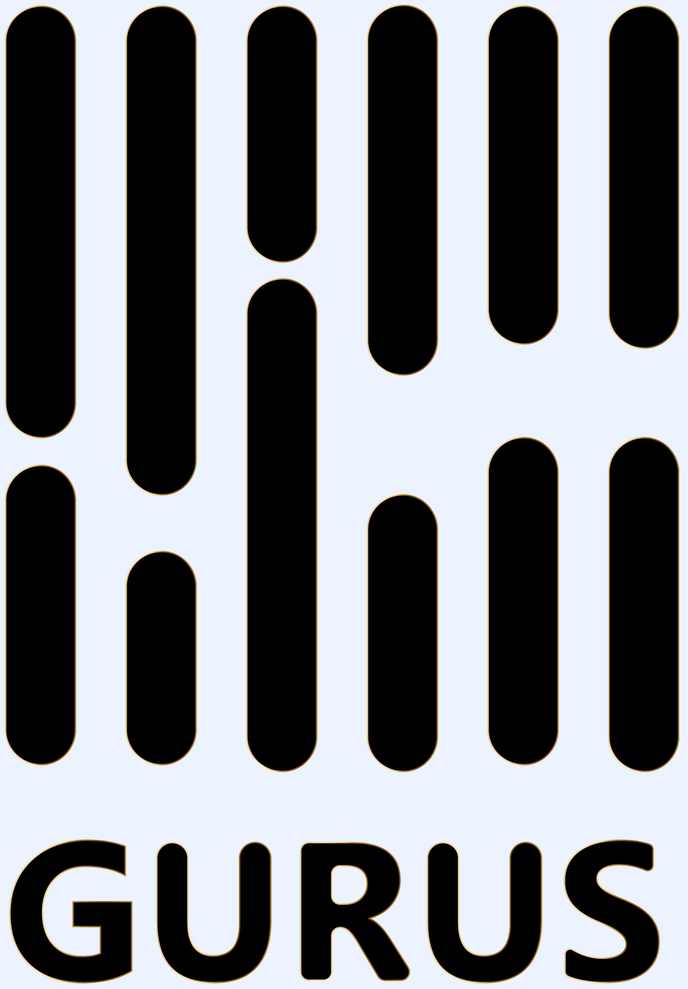

GURUS

GURUS ?

SO WHAT?

A GURU IS A SANSKRIT TERM FOR A "MASTER OR TEACHER" GURU MEANS LITERALLY DISPELLER OF DARKNESS WE TRY TO SHED LIGHT ON DATA TO REVEAL ITS MEANING

OUR INSPIRATION

DATA CAN BE FOUND EVERYWHERE, BUT IT'S HARD TO KNOW THE STORY BEHIND IT AND WHAT IT CAN PRODUCE, GURUS ARE THE ONES THAT CAN EXTRACT KNOWLEDGE FROM DATA AND USE IT TO DO BETTER IN THIS WORLD.

WE ARE GURUS

OUR COMPANY STORY

GURUS PAST

6 different people with
6 defferent specialties
in IT

GURUS PRESENT

An entity that it's sole
goal is to solve a
problem in Ibn Khaldoun
University

GURUS FUTURE

Our Company will
evolve to involve solving
problems outside of Ibn
Khaldoun University

WHAT IS GURUS?

What We Believe In

TRANSPARENCY

RELIABILITY

CONSISTENCY

RESPONSIBILITY

Using Our Logo

ONLINE

IN PRINT

OTHER APPLICATION

Why: Your Reason for Being

What problem do you want to solve?

It is no secret that students in Ibn Khaldoun University have been constently failing and losing interrest in completing their studies, and since there is a lot of data on our students we came up with the idea of predicting students with high rate of Dropping School and try to motivate them using IT, this idea is called HAFEZNY which is the litteral translation of "Motivate Me" from our native tongue which is Arabic

Who: The Pioneer Team

Is your team equipped and well-balanced?

Our team is consistant 6 Core memebers that are specialized in different fields, such as Data Sience, UX Design, Project Managment, Web Development,Software Development Modeling and Mobile Development

Meet the Team

GET TO KNOW US



ALI LEBANI



YOUCEF KHARROUBI



MOSTAFA LAOUMIR



ABDELHAKIM AZZOUZ



LAHCEN RAHLAOUI



NOUAR CHAWKI

GURUS AND HAFEZNI

HAFEZNi a system that connects Students and our Pedagogical on another level of understanding the actual needs of our students and making it easier for our Professors to provide.

HAFEZNI

BRAND

COLOR PALETTE

#77C0C0

#A4D080

BRIEF

As shown our logo has a shape of a staircase representing students paths and a crown at the top that represents sucess, our choice of colors also has a meaning that the paths gets clearer and lighter as you succeed

LOGO OF SYSTEM

Work Plan



The Middle of every succesful project looks like a disaster

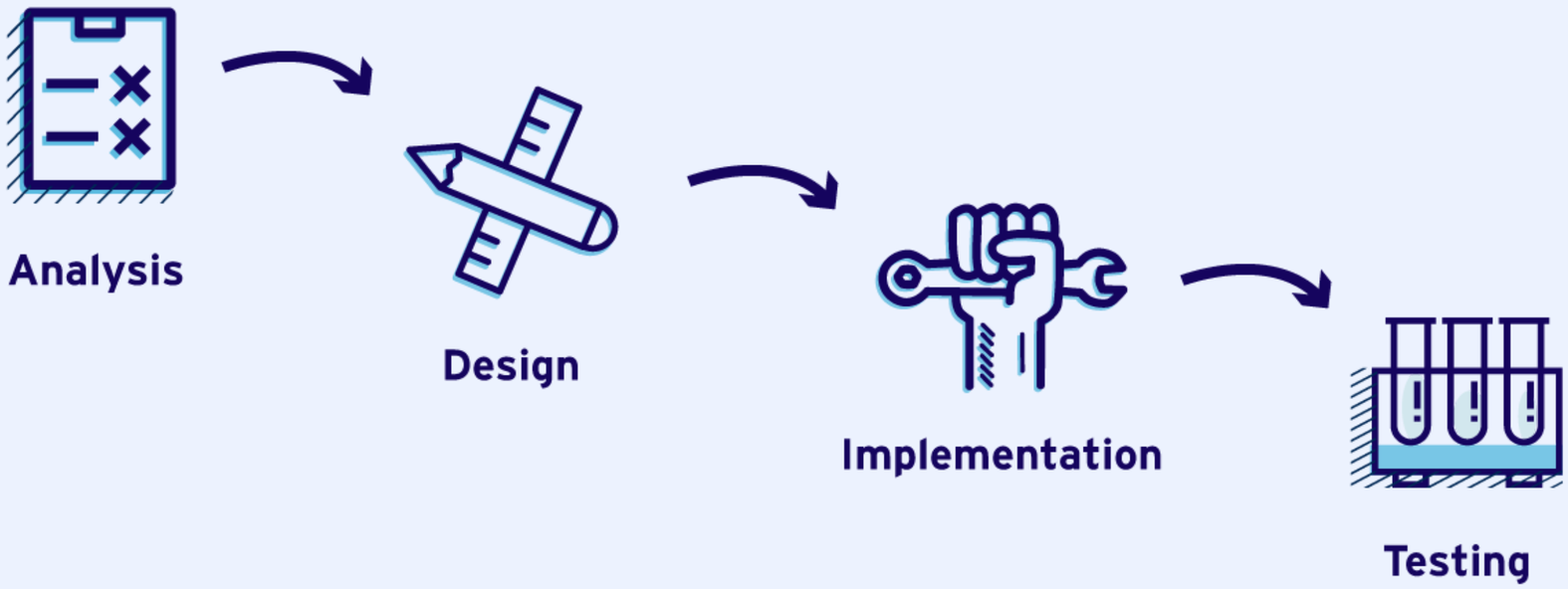
by rosabeth moss kanter



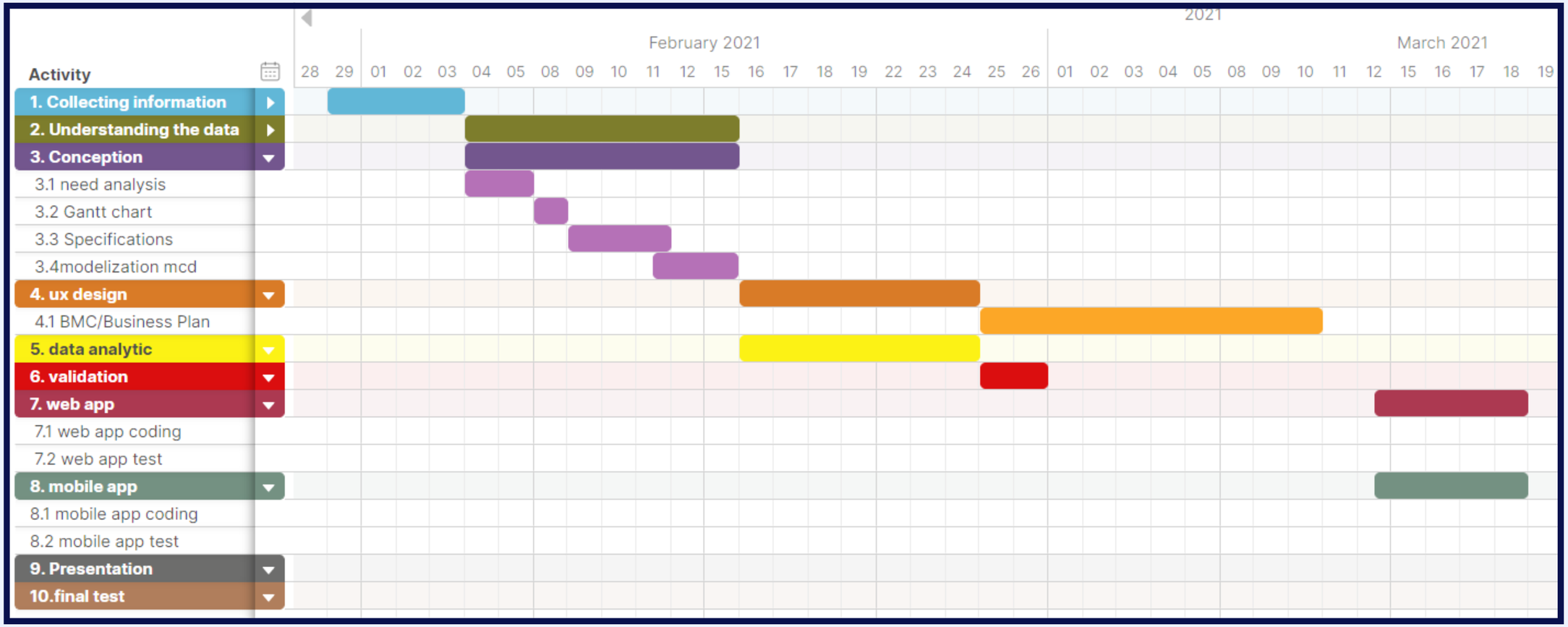
Methodology

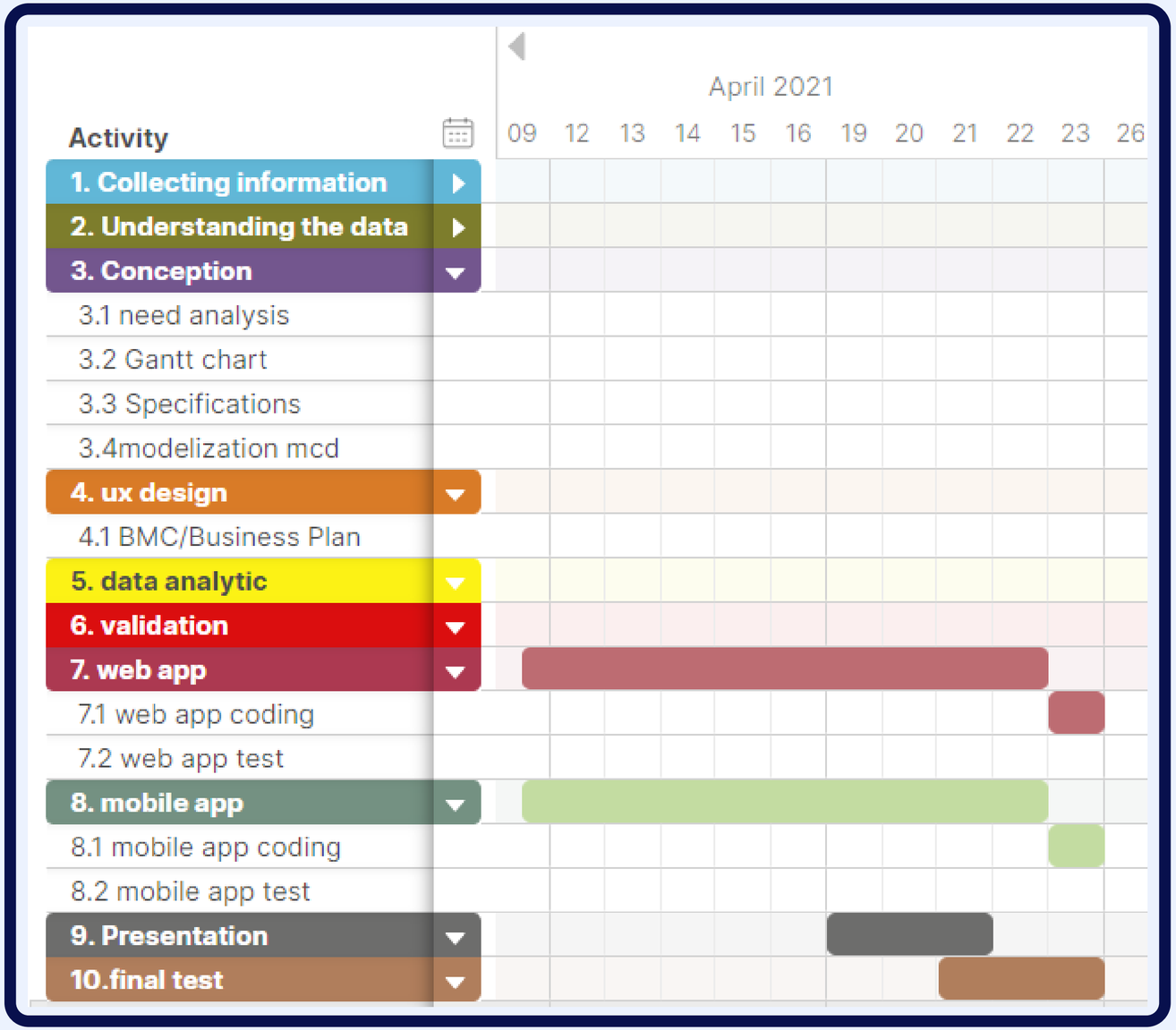
Waterfall Model

Waterfall Method : an example of the Sequential model for managing software development.



Planification




April 2021
Activity
09
12
13
14
15
16
19
20
21
22
23
26
1. Collecting information
2. Understanding the data
3. Conception
3.1 need analysis
3.2 Gantt chart
3.3 Specifications
3.4modelization mcd
4. ux design
4.1 BMC/Business Plan
5. data analytic
6. validation
7. web app
7.1 web app coding
7.2 web app test
8. mobile app
8.1 mobile app coding
8.2 mobile app test
9. Presentation
10.final test

Specification

Users



Student

A Student will be able to be kept in touch and be close to his schedule, his Scholar Status and his Professors



Academy

The Academy will have a close and detailed Status on its Students individually and as a whole Unit, and be able to motivate them through sending Notif

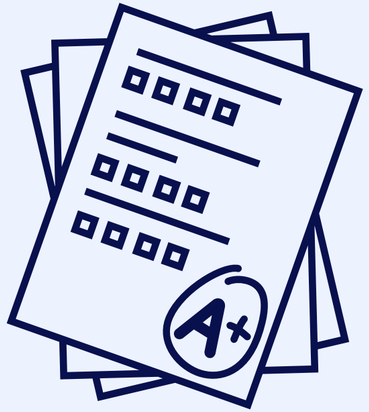
Requirements

functionnal

our system must help to detect students with high rates of Dropping School acording to their

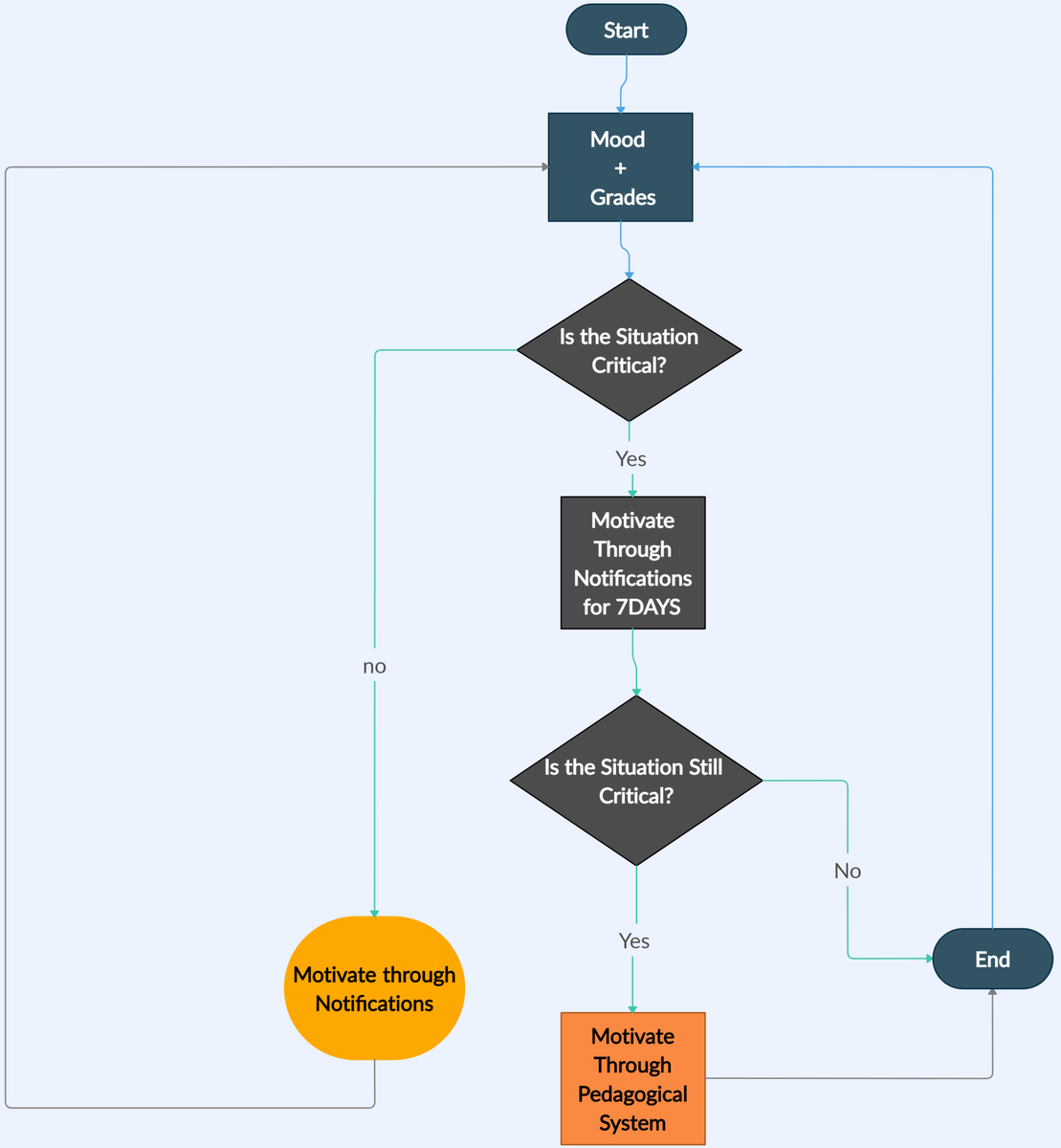


Mood



Test Scores

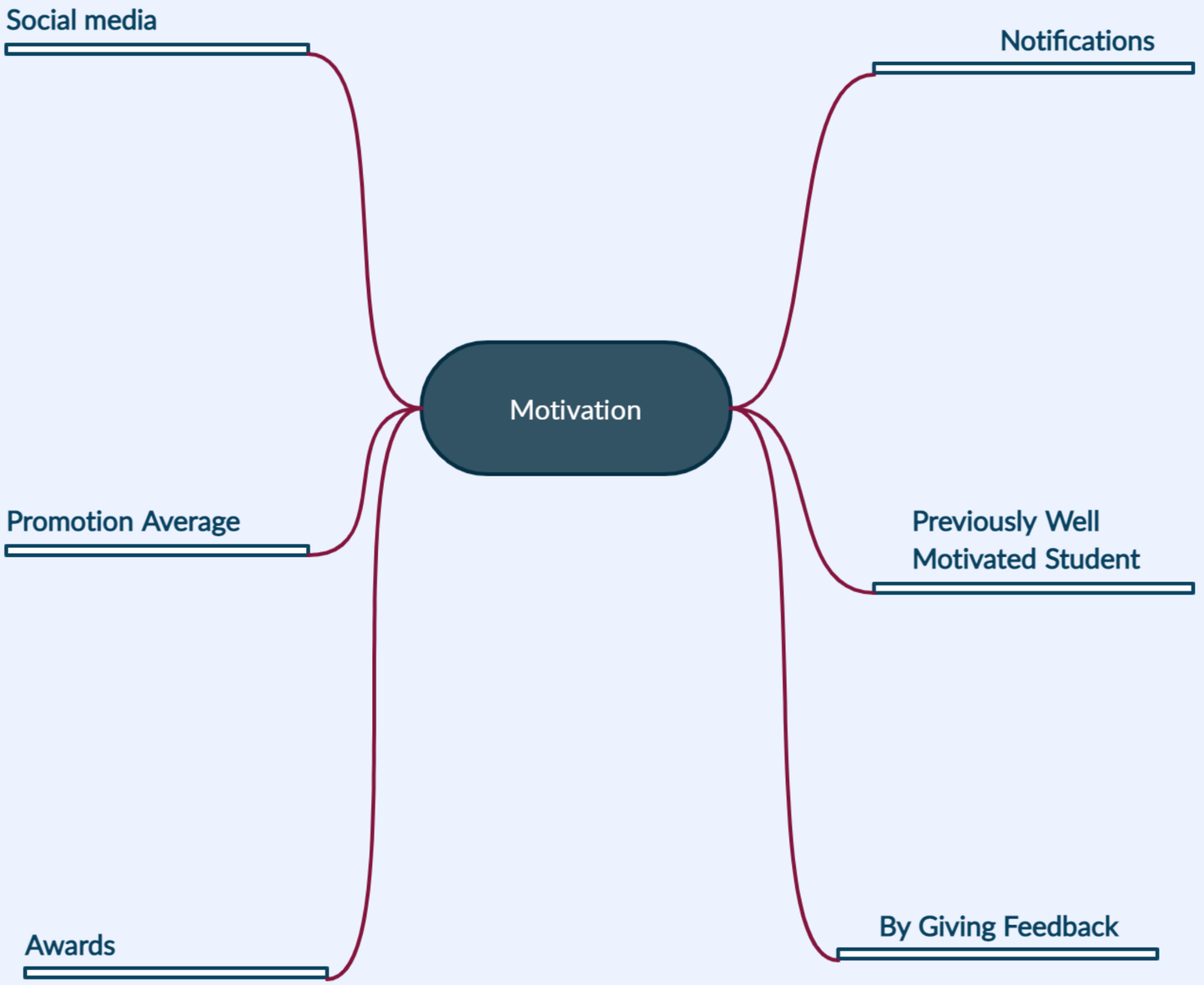
We Used two important features that help us detect if a student is Unmotivated and those features are **Mood**, and **Grades**

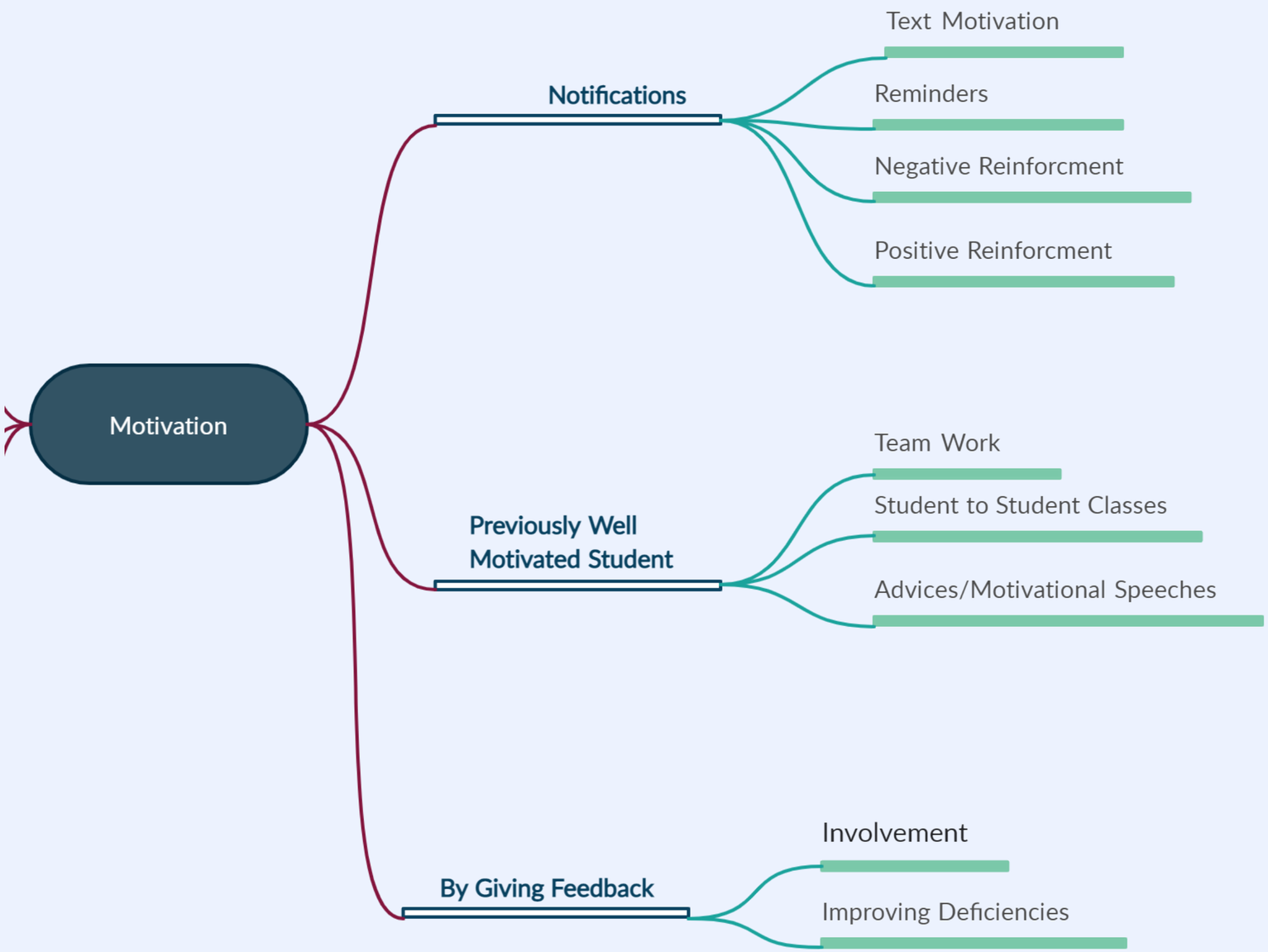

Start
Mood
+
Grades
Is the Situation Critical?
Yes
Motivate Through Notifications for 7DAYS
no
Is the Situation Still Critical?
No
Yes
Motivate through Notifications
End
Motivate Through Pedagogical System

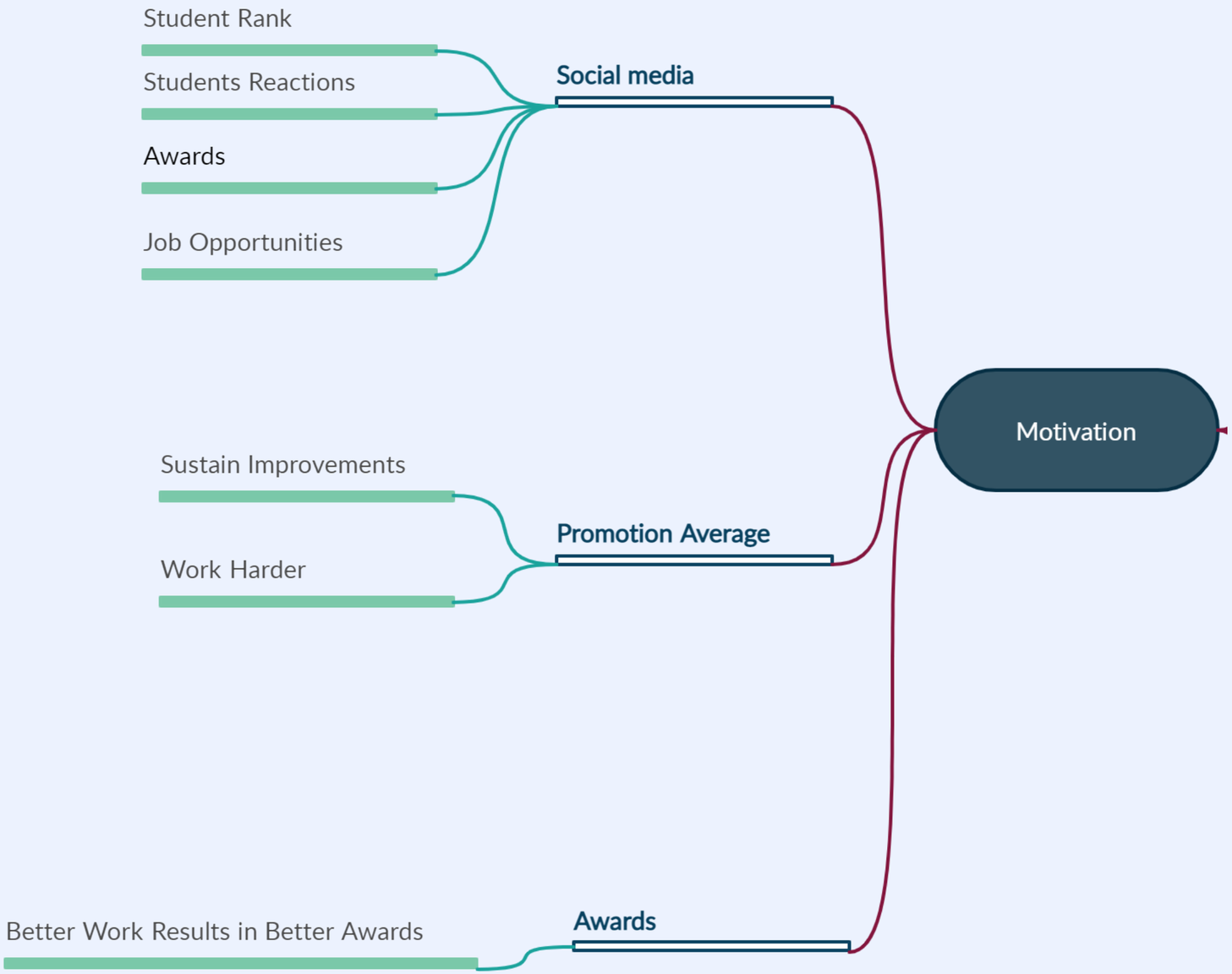
Motivation






Social media
Notifications
Motivation
Promotion Average
Previously Well
Motivated Student
Awards
By Giving Feedback


Motivation
Notifications
Text Motivation
Reminders
Negative Reinforcment
Positive Reinforcment
Previously Well Motivated Student
Team Work
Student to Student Classes
Advices/Motivational Speeches
By Giving Feedback
Involvement
Improving Deficiencies


Student Rank
Students Reactions
Awards
Job Opportunities
Social media
Motivation
Sustain Improvements
Work Harder
Promotion Average
Better Work Results in Better Awards
Awards

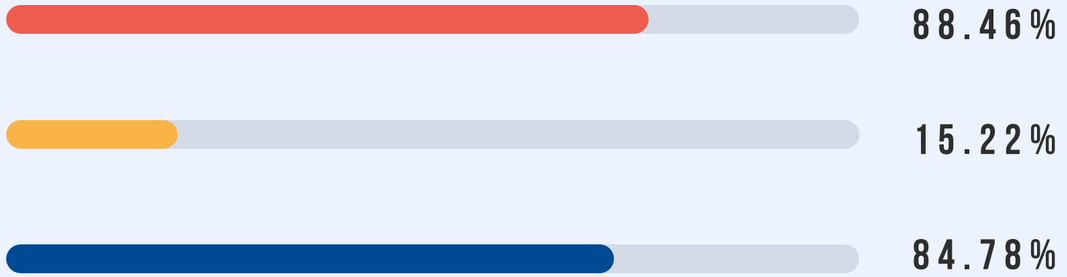
UX research

USER INTERVIEW

01 46 Out of 52 Of Our Students Use Smartphones

02 7 Out of 46 Phone Users Use IOS

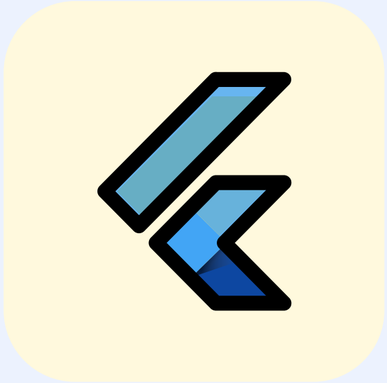
03 39 Out of 46 Phone Users Use ANDROID



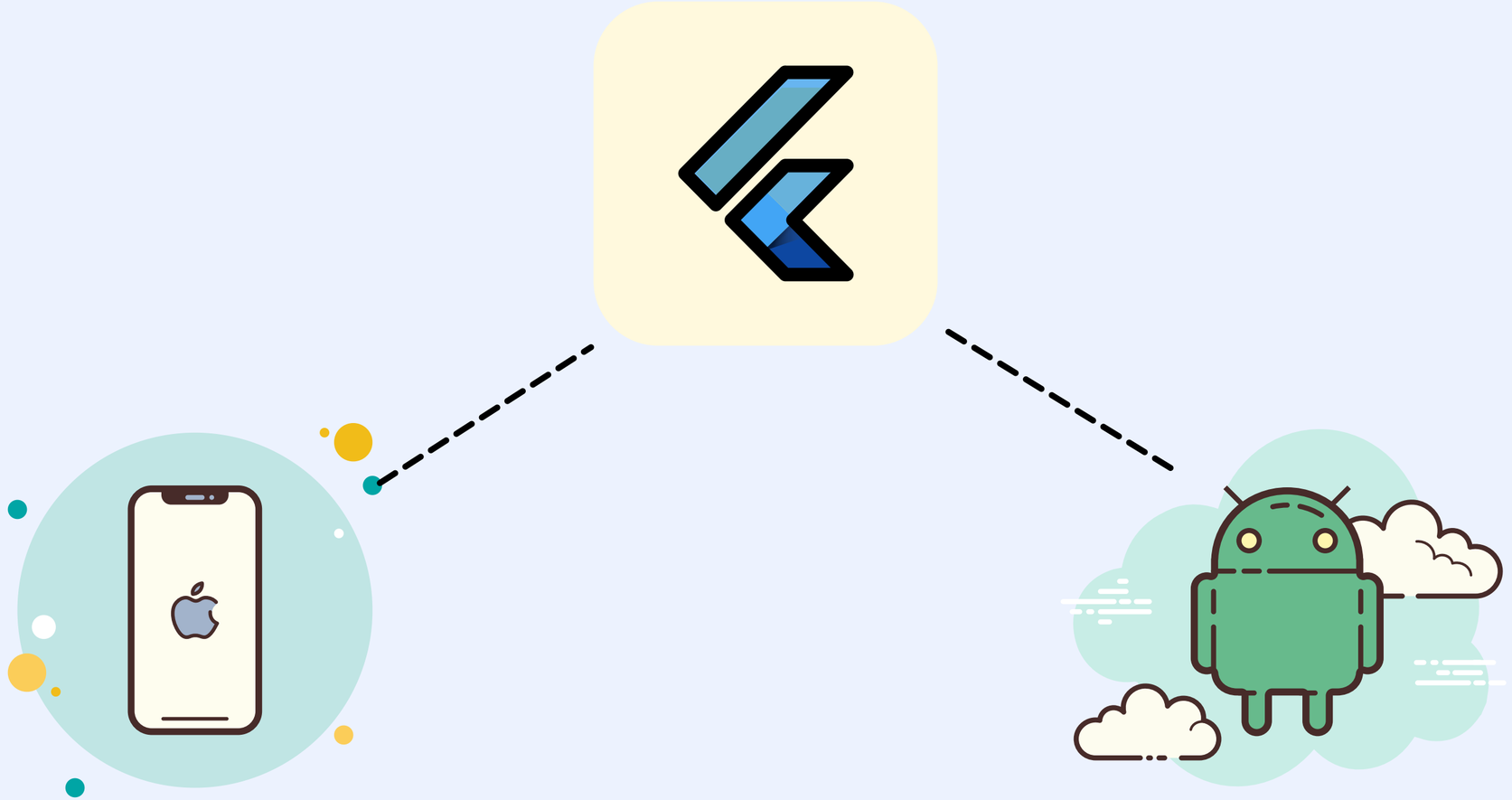
01 40 Out of 52 Phone Users Prefer DARK MODE

02 12 Out of 52 Phone Users Prefer LIGHT MODE





Flutter is an open-source UI software development kit created by Google. It is used to develop applications for Android, iOS



Performed by google and apple, dark mode became one of the most requested features and a super essential part of UI



Mobile wireframing
HAFEZNI

E DATA

Data Analytics

. **Tools**

. **Data Warehouse implementation**

- . Extract, Transform, Load
- . Multidimensional modeling
- . Cubes implementation
- . OLAP queries
- . Knowledge extraction and decision making
- . Predictions

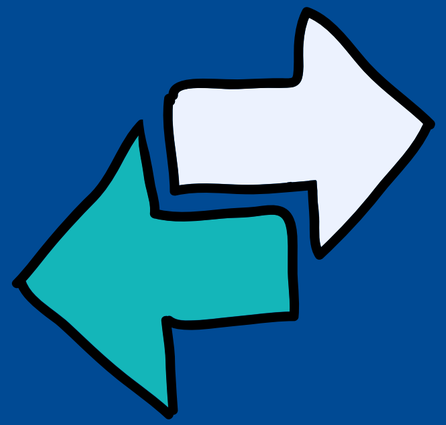
. **Problems encountered**

- . Data quality
- . Unbalanced Data

Data Warehouse Implementation



- **Extract**
  - **Extractions of data from given sources**
    - **CSV files**
      - Data about the first-year student
      - Data about the master degree Student
    - **Mobile Application**
- **Extractions done by Talend data integration tool**

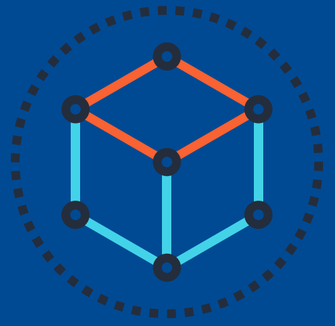


. **Transform**

. **handle missing values**

. handle incorrect and inaccurate data

. create new meaningful columns

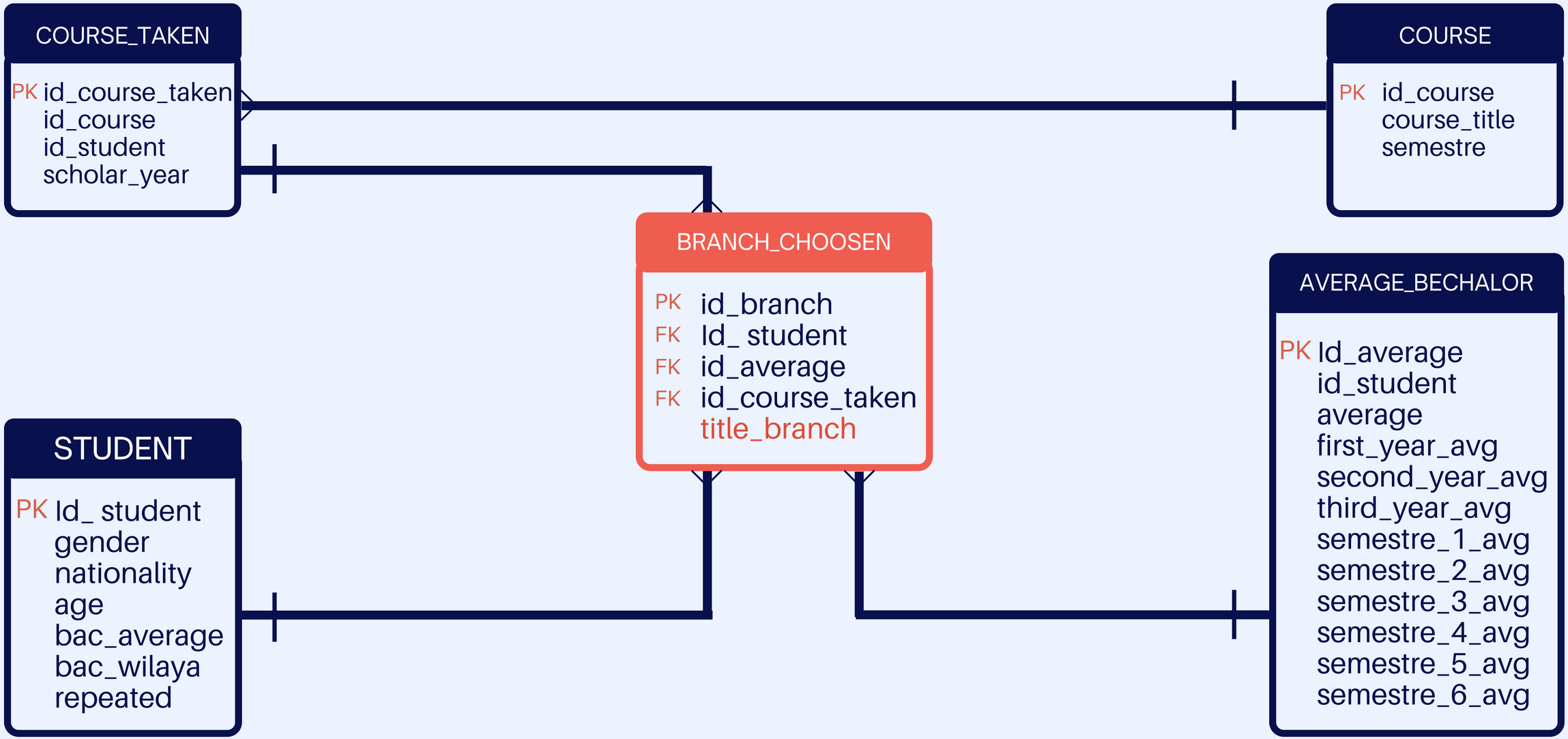


. Multidimensional modeling

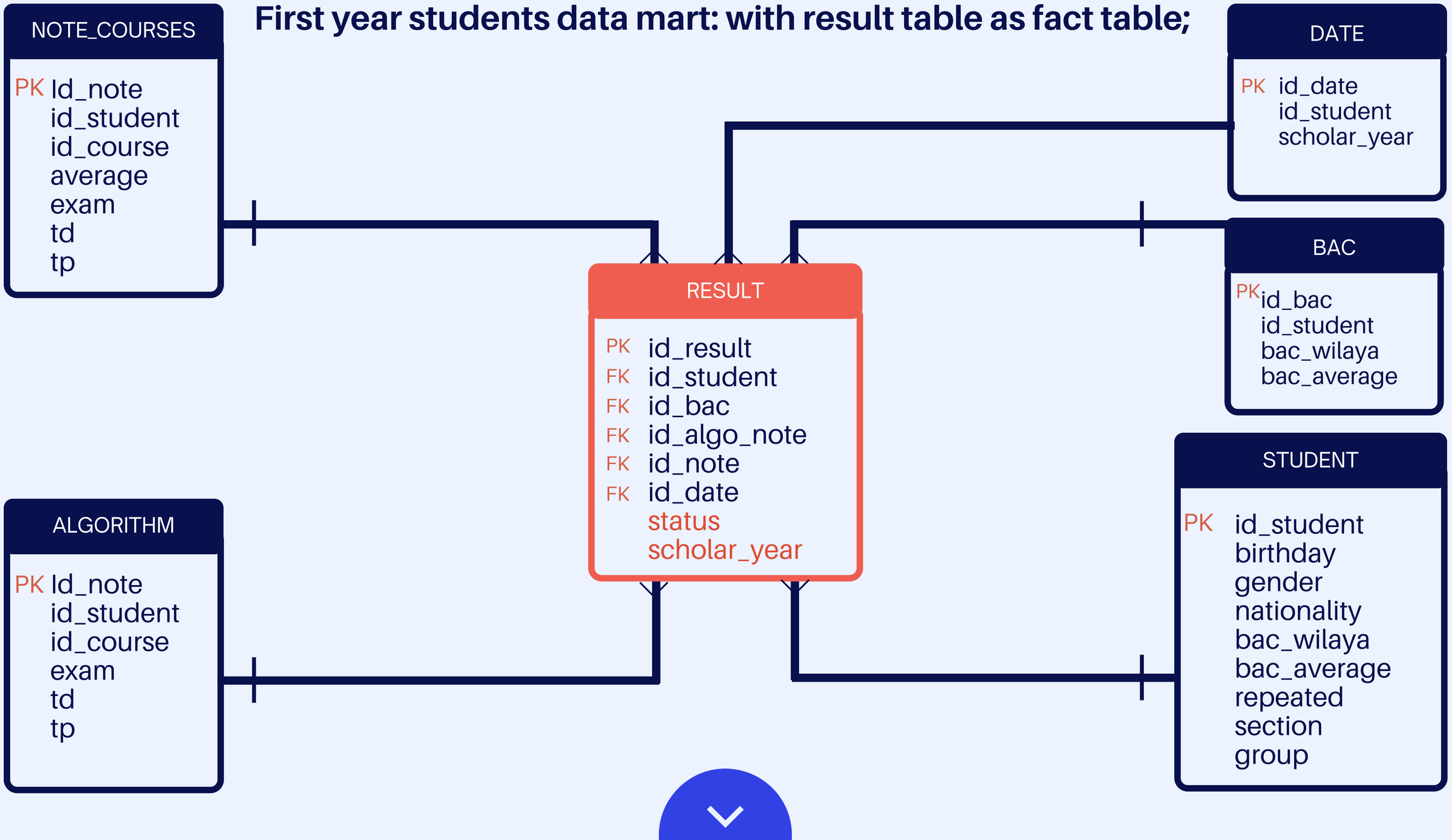
. After consideration and study for business objectives

. we defined two data marts:

Master degree: with branch_chosen table as fact table



First year students data mart: with result table as fact table;



Cubes implementation

Use SQL to select the variables to analyse bepending on business needs

```sql
-- result cube query

SELECT
    b.bac_average,
    d.scholar_year,
    s.repeated,
    r.status

FROM result r

JOIN bac b  ON r.id_bac  = b.id_bac
JOIN date d ON r.id_date = d.id_date
JOIN student s ON r.id_student = s.id_student

GROUP BY d.scholar_year;

-- LIMIT 20
;
```

```sql
-- branch cube

SELECT
    a.average,
    c.id_course,
    s.bac_average,
    b.title_branch

FROM branch b

JOIN courses_taken c    ON b.id_course_taken  = c.id_course_taken
JOIN average a          ON b.id_moyenne = a.id_moyenne
JOIN student s          ON b.id_student = s.id_student

GROUP BY c.id_course
```

OLAP queries
Dicing

```sql
-- result dice query

SELECT
    b.bac_average,
    d.scholar_year,
    s.repeated,
    r.status

FROM result r

JOIN bac b  ON r.id_bac  = b.id_bac
JOIN date d ON r.id_date = d.id_date
JOIN student s ON r.id_student = s.id_student

WHERE   d.scholar_year IN ('2010', '2011')
AND     (b.bac_average < 11 AND b.bac_average >= 10)
AND     s.repeated = 'false'

GROUP BY d.scholar_year

-- LIMIT 20
;
```

```sql
SELECT
    a.average,
    c.id_course,
    s.bac_average,
    b.title_branch

FROM branch b

JOIN courses_taken c    ON b.id_course_taken  = c.id_course_taken
JOIN average a          ON b.id_moyenne = a.id_moyenne
JOIN student s          ON b.id_student = s.id_student

WHERE   c.id_course IN ('Agl', 'XMLW')
AND     (s.bac_average < 11 AND s.bac_average >= 10)
AND     (a.average < 11 AND a.average >= 10)

GROUP BY c.id_course
```



. OLAP queries
. Dicing
. Slicing
. Roll-up
. Drill-down



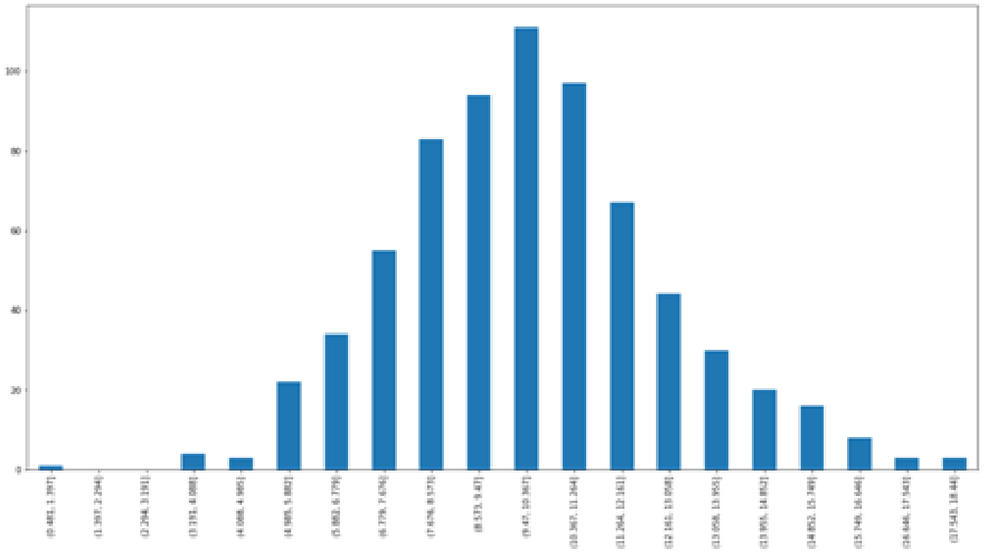
Knowledge extraction and Prediction

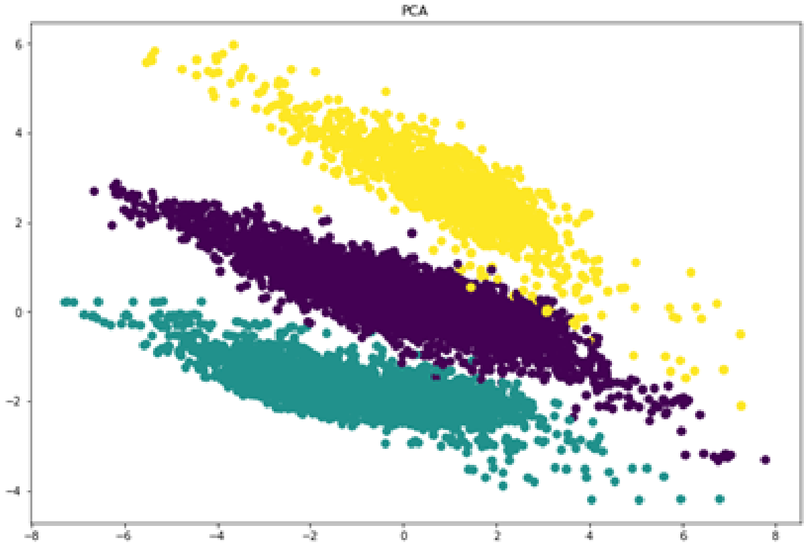
. Knowledge extraction and Prediction

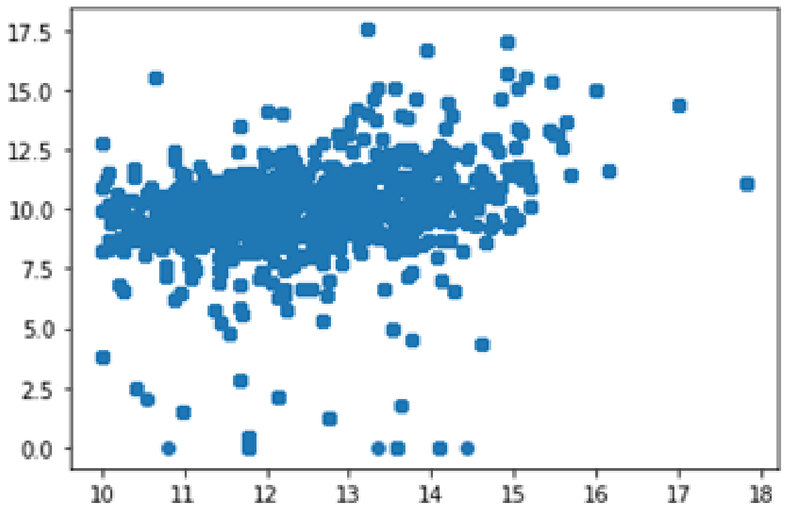
. Determine whether a student will be admitted or will be adjourned based on his grades during his studies

. Find correlation between different variables and columns that will help in decision making process.

. This is done using different data analysis and visualization techniques as well as various prediction models.




PCA


17.5
15.0
12.5
10.0
7.5
5.0
2.5
0.0
10
11
12
13
14
15
16
17
18

Problems encountered

- Data quality
- Unbalanced Data
- Undocumented Data

Models	Accuracy
SVM	95%
logistic Regression	94%
KNN	82%
DEEP Neural Networks	91%

Business Plan & Business Model

Set up for success with strategic planning frameworks, including SWOT Analysis, and more.



Create a Business Model

Key Partnerships

friendly Products and Companies

Key Activities

Content Update
Human Resourcr for Gurus
UX reasearch
Design for student
Psychological Expert Assistance

Key Resources

Development
Designing
Psychology

Value Propositions

Direct Publicity and marketing tool to customers Reminder
Better data + Data visualisation on students.
Motivate UnMotivated Students OnDemand

Customer Relationships

Prizes and Recognition
Hafezni + Gurus Community event.

Channels

Play Store or appstore for students online.
website for a competent authority of university officials.

Customer Segments

-Direct Customers "Students".
-Bussiness"University"

Cost Structure

Developement Marketing Prizes "Marchandise "

Revenue Streams

Yearly Subscription "100 da " PartnerShips and Sponsors
Message Marchandise for motivation "T shirt +Laptop .."

Summary

The "Gurus" is team of students from the University of Ibn khaldon working on the project of hafezni seeking to motivate students to study all round the world starting from Algeria by knowing there needs and abilities.

The company and the work team

1 - the marketing name " GURUS"
2 - The future vision of the project - A platform contained in the university structure
3 - the mission of the company
- Motivating students
- Increasing productivity and quality of projects
- Program directed at the university community
4- values and principles : - the fiability - Respect - honesty

5 - targets of the project :

- Short-term goals: - Make the administration work easier - Motivate the student - Update the student with new developments
- Long-term goals: - Create a university community.

6- the customers: - Covering the University of Tiaret (first year)

- Covering the Algerian University (after three years)
- Expanding to a larger segment (after seven years)

7-Projected budget : - Not Found - Work for free

8- Company structure : - The platform is affiliated with the university structure

- Technical - pedagogy - A committee appointed by the Ministry of Higher Education and Scientific Research.

Service or product

Competitive advantages :
-Not having any competitor.
-Symbolic price.
-Efficient work team

Customers

-Motivate the student.
-Knowing the student’s needs and how to meet them.

Market

1. There is no competitor when entering the market, But the first competitors are colleagues.
2. The market does not absorb any competitors because it requires the approval of the higher authorities.
3. Barriers to entry to the :
3.1. Failure to adapt to the idea.
3.2. Rejection of the higher authorities of the idea.

4. Registration fees as 100DA.
5. Providing the service on Android and IOS, downloadable from Appstore, PlayStore and The official website.
6. Promote the product through the university and social media.
7. The expected sales :
    -Worst expectation : 200000000.00 DA
    -Best expectation : 250000000.00 DA



SWOT Analysis

A SWOT analysis defines our company's path toward our goals by identifying strengths and weaknesses (internal features) and opportunities and threats (external conditions).

Strengths

- we don't only use existing data,
  we create new one that will help our project in the future

Weaknesses

Not fit for purpose.

Opportunities

- empty market
- Being bought By "University" Using Direct Student.
- Social Media To promote HAFEZNI

Threats

- Could Develop interest in the opportunity.
- Cloning
- Governmental detention procedures.

Financial plan

The income will be provided from the students 100 DA for each student a year.
the income will be split into 2 parts [X%] for the government what would provide all the needs for the project and [100%-x%] For the team from this part 70% will be dedicate to research on the development and 30% for the founders.

Tools List

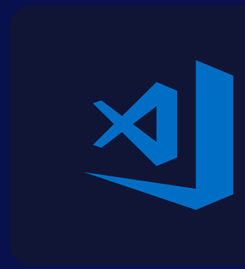
Here is A list of the tools that helped us get our work done smoothly













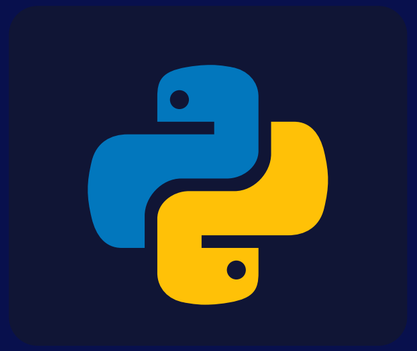









TOOLS LIST

















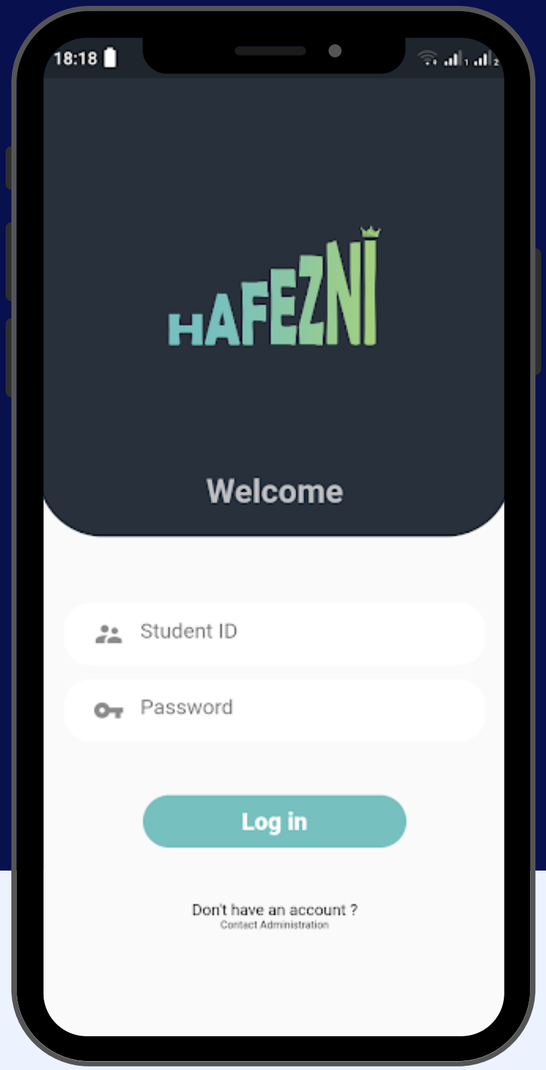
PROGRAMING LANGUAGE

Mobile APP Project



SPLASH SCREEN

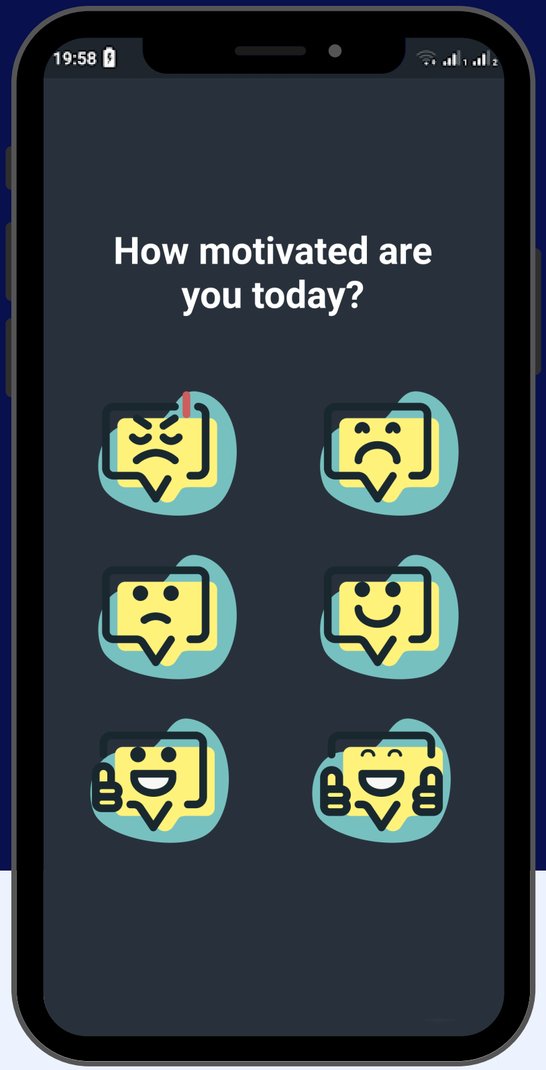
A screen with HAFEZNI logo and our company's Watermark that sits for 3 seconds as the rest of the application Loads in the Background



LOGIN PLATFORM

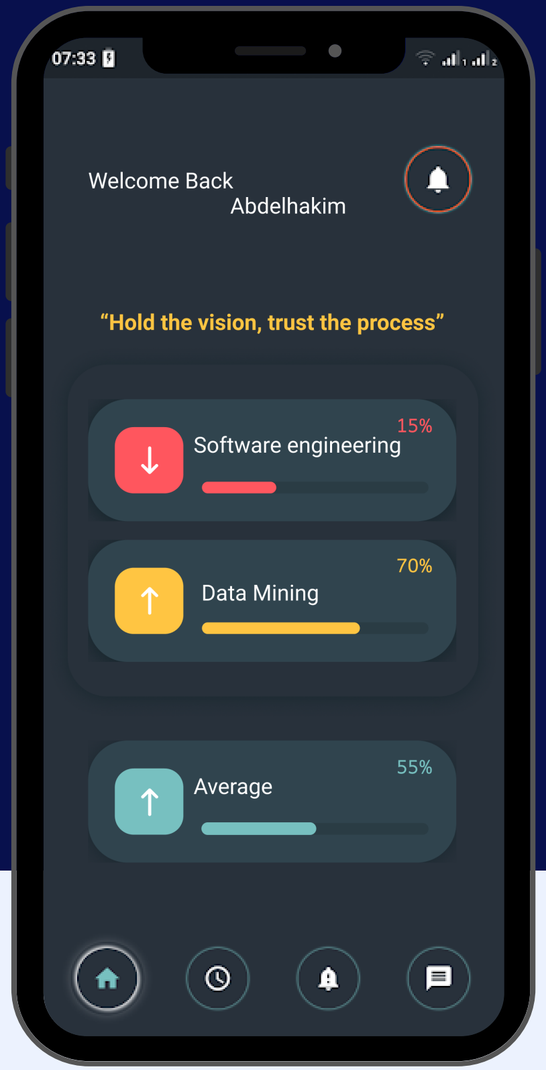
At this Platform you have to insert SudentID and a Password to be able to proceed to the rest of the Application

MOBILE PROJECT



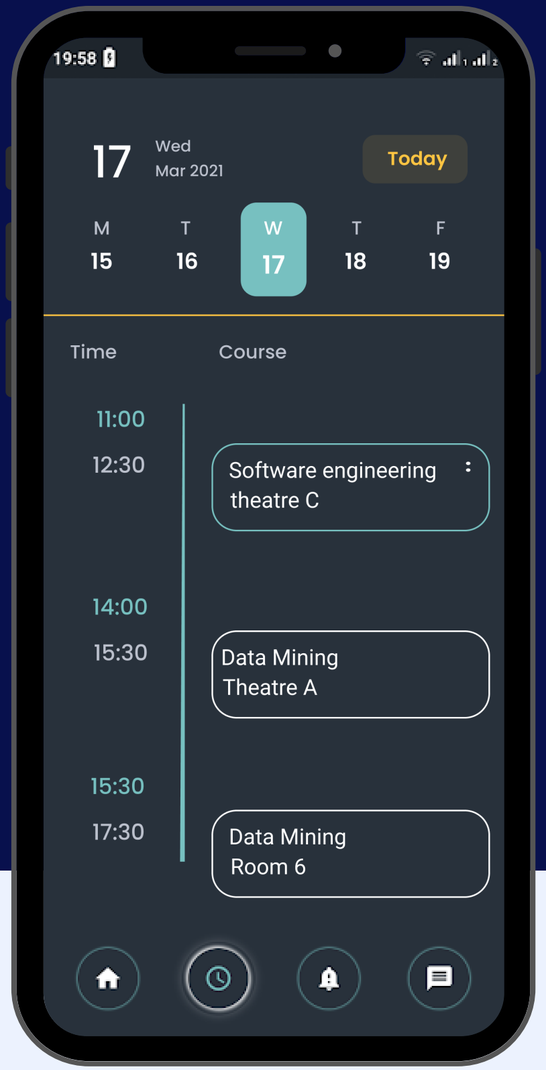
SELF REPORTING

This Activity lets the Student self report his motivation level that will be sent to the Database later for more precised Predictions



MAIN SCREEN

This Activity shows you how good you are doing at your classes with the average of the whole Promotion to keep you focused and in touch with your grades and to see how good you are doing compaired to the whole Promotion



SCHEDULE

This Activity helps you see what classes you have today and lets you check your classes for the whole week

MOBILE PROJECT



ANNOUNCMENTS

You could see at the top-right of the Main Activity there is a bell with a changing color that changes if you get a new Announcment

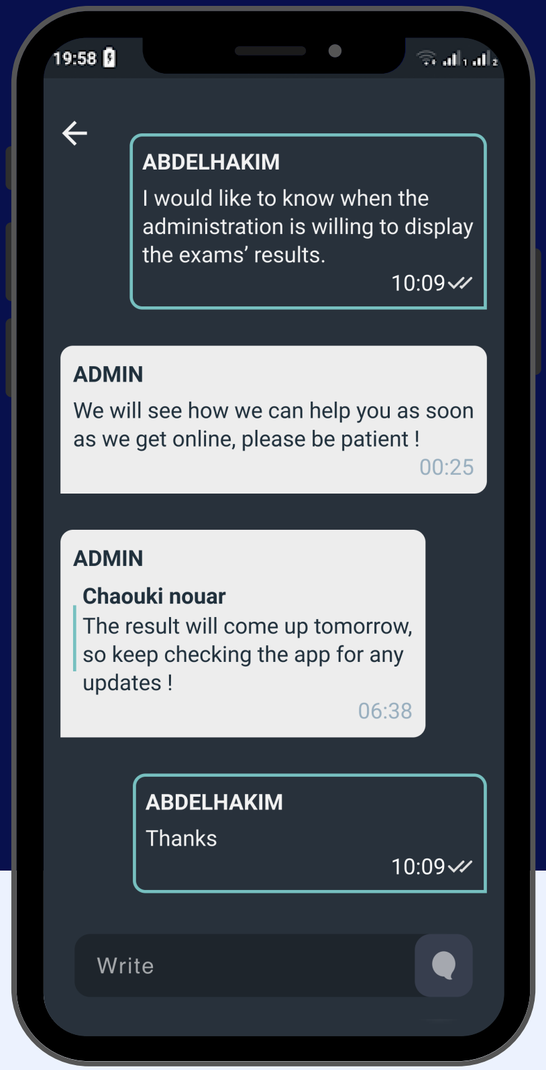


Congatulations now
Hi Abdelhakim!
Congrats, you have won a T-shirt, plus an online cours. proceen to this link to claim your gifts.
https://hafezni.dz/prise/UJK94dy78kl

ANNOUNCMENTS

at this activity you could see your announcments, which can be either a Reward, Motivationnal Activity or an announcment

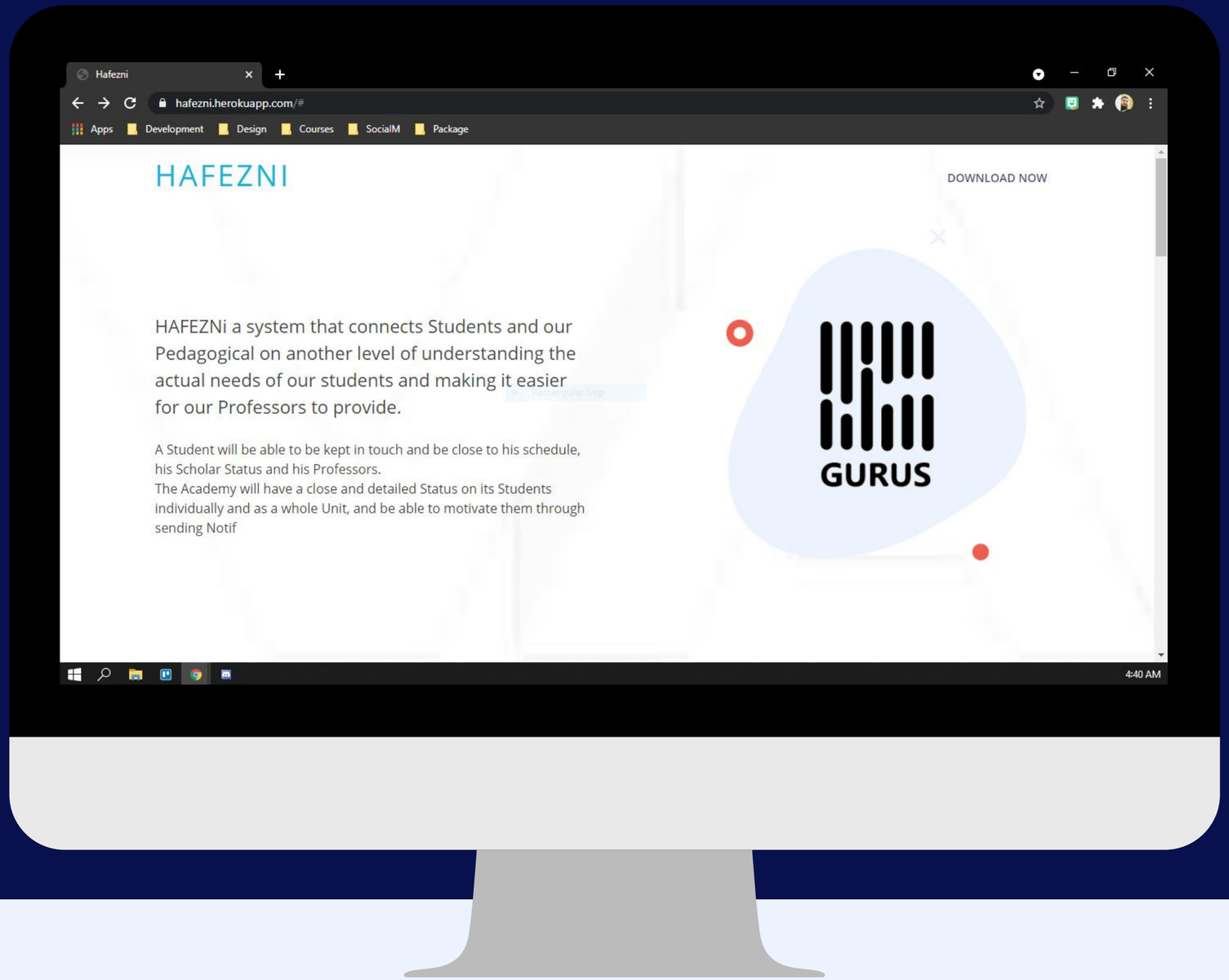
MOBILE PROJECT



CONTACT

This Activity lets you Contact the Academy for any kind of help or questions that you have

Mobile App Description WebPage



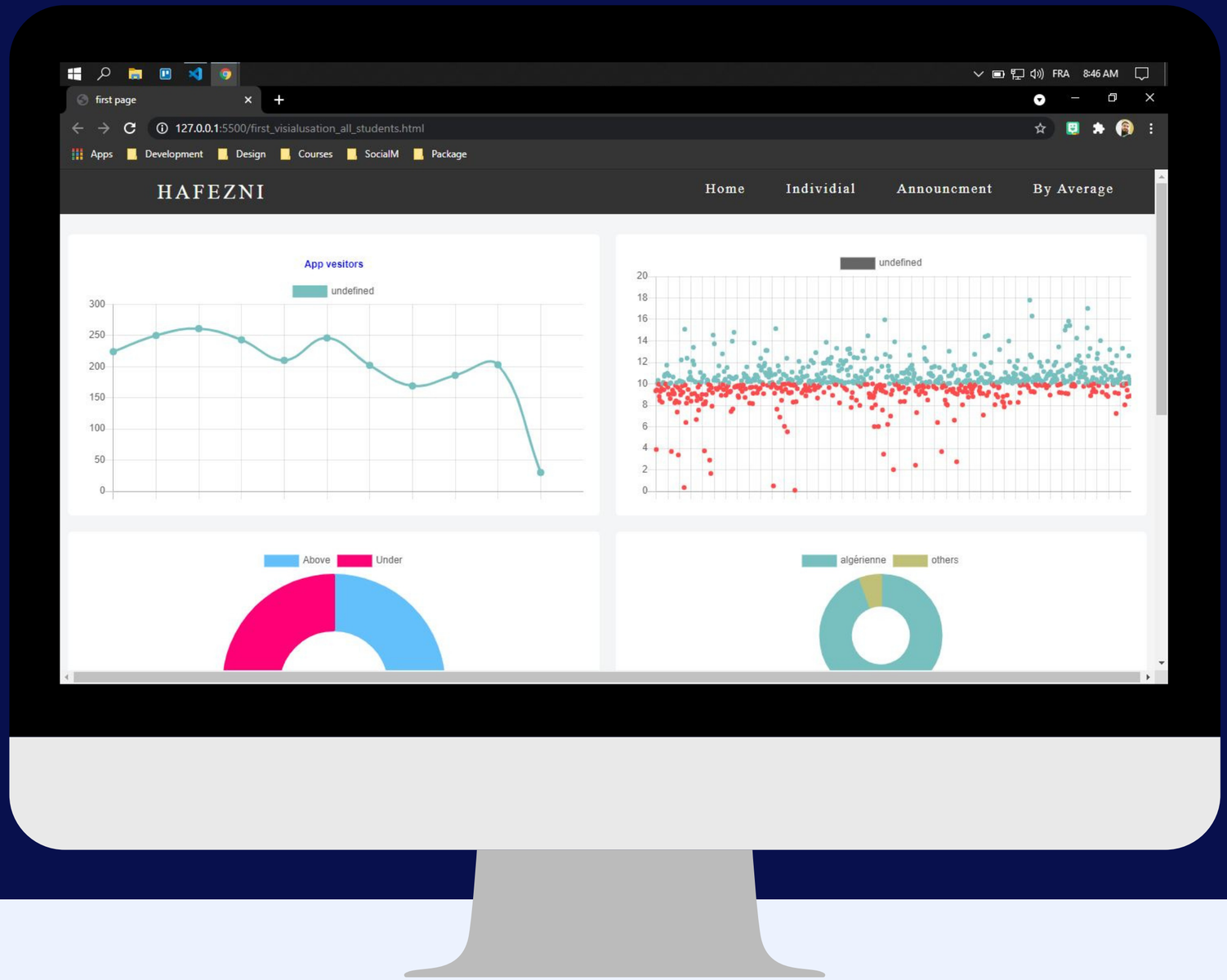
MOBILE APP DESCRIPTION WEBPAGE

this web page is a discription to our Mobile app where you can read a brief description about it, about our team and how to download it.

https://hafezni.herokuapp.com/

WEBPAGE PROJECT

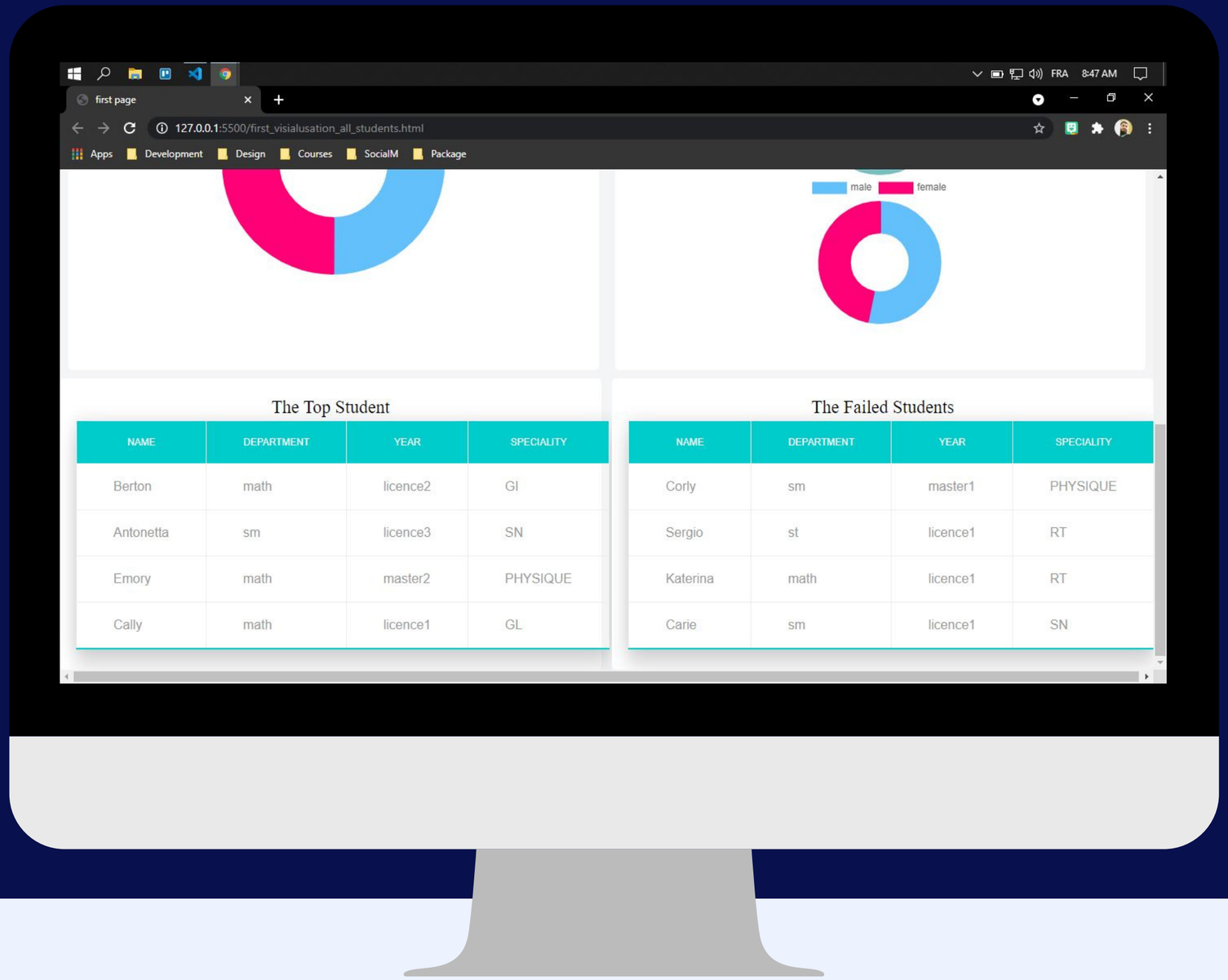
Web app project



WEBSITE FOR ACADEMY

This website is dedicated to help our pedagogical systemhave a specific and detailed view on students data using charts, and react to it easily

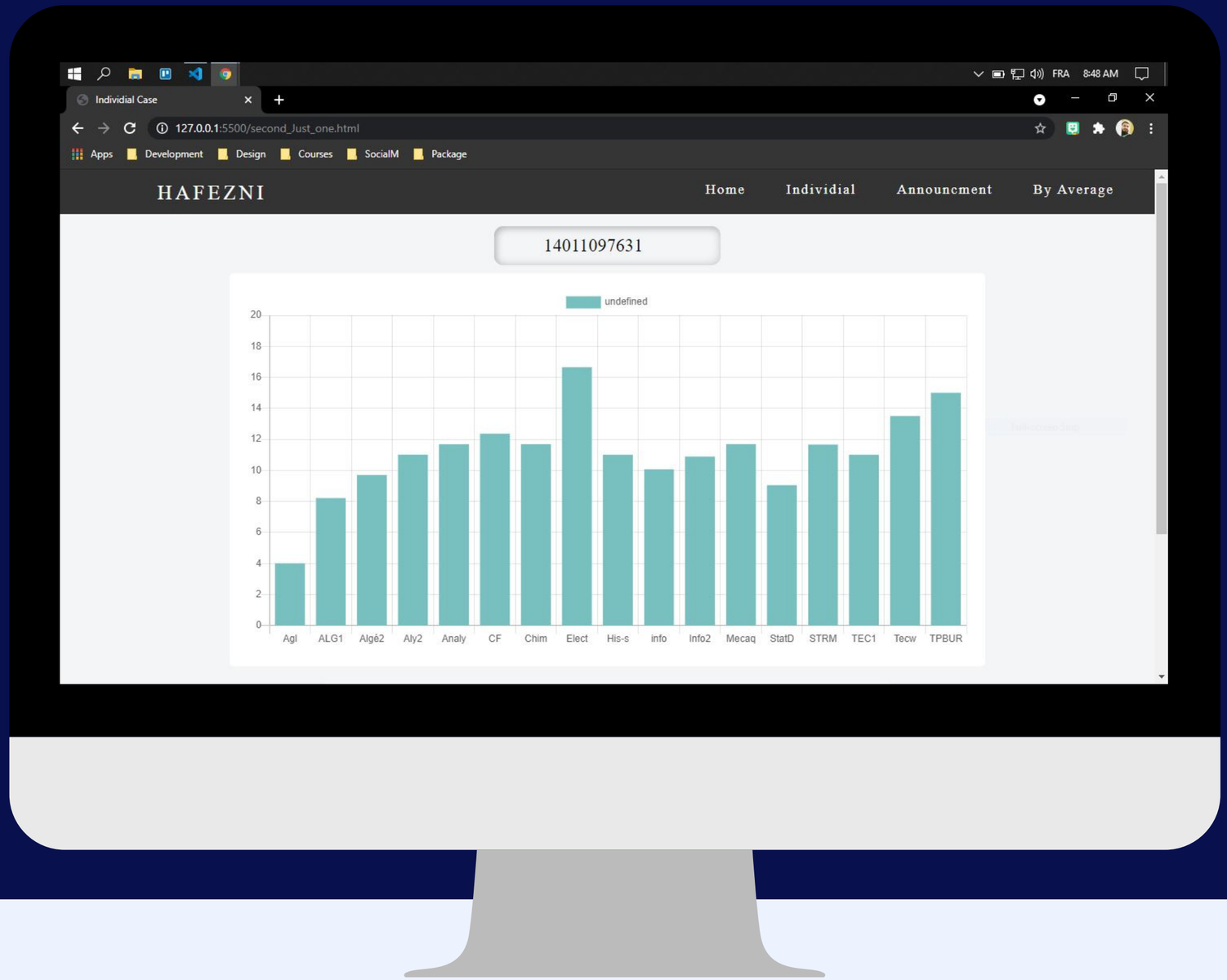
WEBPAGE PROJECT



WEBSITE FOR ACADEMY

This web page gives a good detailed data about failing students and succeding students

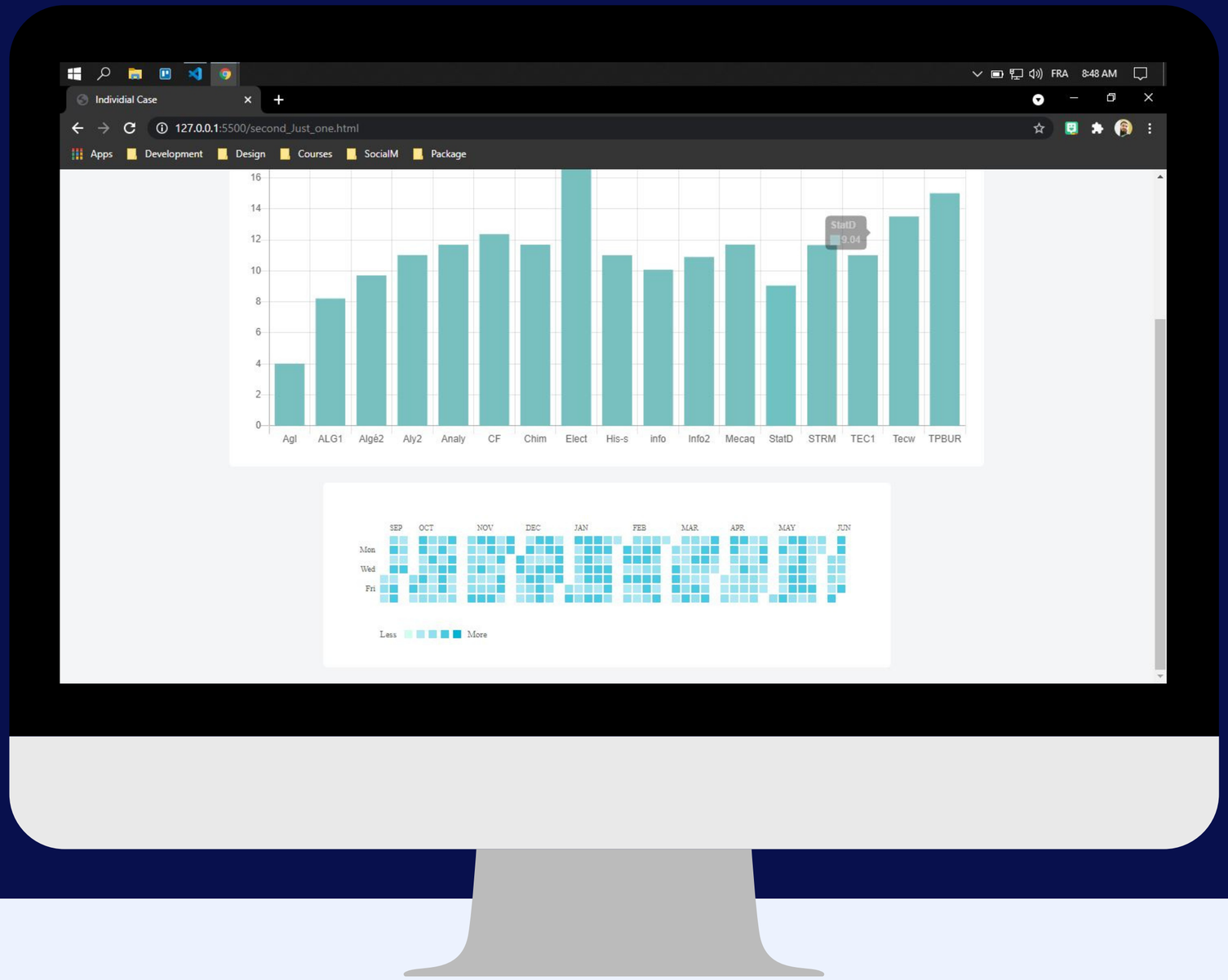
WEBPAGE PROJECT



WEBSITE FOR ACADEMY

this web page shows bar charts about classes for each student, that can be found by their ID

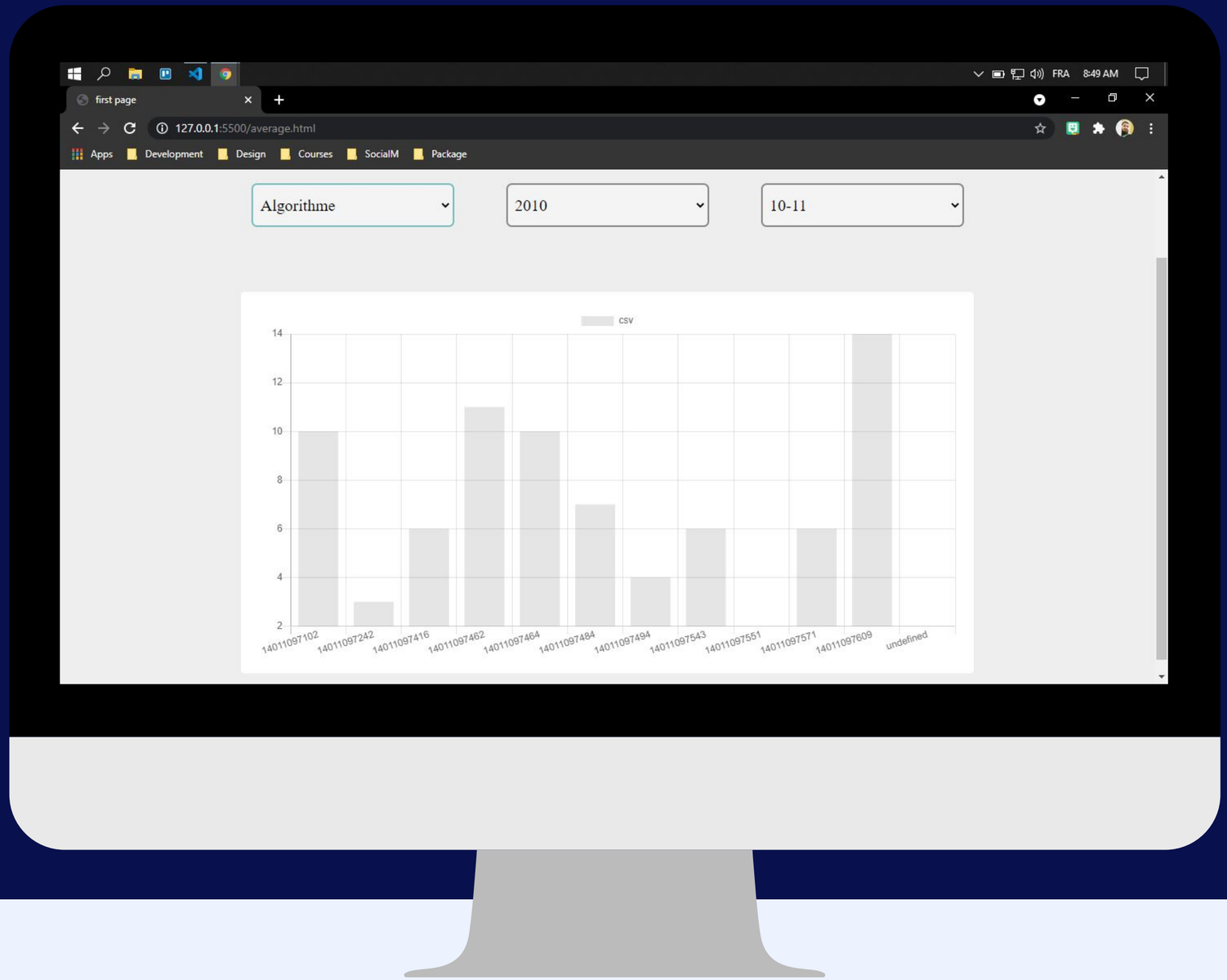
WEBPAGE PROJECT



WEBSITE FOR ACADEMY

this web page shows heat map chart about classes for each student, that can be found by their ID

WEBPAGE PROJECT



WEBSITE FOR ACADEMY

this web page shows bar charts about classes for a given year for all students filtered by their average, and it can also send notifications to students

WEBPAGE PROJECT

We're done!

